Chapter 16 Homework Answer Sheet

Part A Multiple choice (15 points)

1-5 CBADB

6-10 ABECD

11-15 BEEED

Part B Short questions (15 points)

16. (12 points)

(a) (2 Pts) $HCOOH(aq) + H_2O(l) \rightleftharpoons HCOO^-(aq) + H_3O^+(aq)$

or $HCOOH(aq) \rightleftharpoons HCOO^-(aq) + H^+(aq)$

$$K_a = \frac{[H_3O^+][HCOO^-]}{[HCOOH]} \quad \textbf{or} \quad K_a = \frac{[H^+][HCOO^-]}{[HCOOH]}$$

(b) (3 Pts)

$pK_a = 3.74 \quad K_a = 10^{-3.74} = 1.82 \times 10^{-4} = 1.8 \times 10^{-4}$

	$HCOOH(aq)$	$+\ H_2O(l)$	$\rightleftharpoons HCOO^-(aq)$	$+\ H_3O^+(aq)$
Initial concentration / *M*	0.25	-	0	0
Change concentration / *M*	–x	-	+x	+x
Equilibrium concentration / *M*	0.25–x	-	x	x

$$\frac{x^2}{0.25 - x} = 1.82 \times 10^{-4}$$

Assume x << 0.25, then $x^2/0.25 = 1.82 \times 10^{-4}$

$[H_3O^+] = x = 0.00674$ *M*

$pH = -\log[H_3O^+] = -\log(0.00674) = 2.17$

(c) (4 Pts)

$$K_b = \frac{K_w}{K_a} = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-4}} = 5.56 \times 10^{-11} = 5.6 \times 10^{-11}$$

$HCOO^-(aq) + H_2O(l) \rightleftharpoons HCOOH(aq) + OH^-(aq)$

I/*M*	0.20	--	0	0
C/*M*	-x	--	+x	+x
E/*M*	020-x	--	x	x

$$K_b = \frac{[HCOOH][OH^-]}{[HCOO^-]} = \frac{x^2}{0.20 - x} = \frac{x^2}{0.20} = 5.56 \times 10^{-11}$$

$x = [OH^-] = 3.33 \times 10^{-6}$ *M*

$pOH = -\log[OH^-] = 5.478$

$pH = 14.00 - pOH = 14.00 - 5.478 = 8.52$

(d) (3 Pts) The solution with 1 *M* NaH_2PO_4 is acidic.
K_b for hydrolysis of $H_2PO_4^-$ is $K_w / K_{a1} = 1.0 \times 10^{-14} / (7.5 \times 10^{-3}) = 1.3 \times 10^{-12}$, which is smaller than K_{a2} (6.2×10^{-8}) for the dissociation of $H_2PO_4^-$.

17. (3 points)

(a) HY

(b) X^-

(c) The equilibrium will mostly lie to the left ($K_c < 1$).